CTL DECISION SHEET

Standard(s) (incl. year)	Subclause(s)	No.	Year
All standards harmonized with IEC 60664 series and addressing PWBs	General	DSH 2001	2014
Category			
VARIOUS			
Subject	**Keywords**	**Developed by**	**Approved at**
Creepage distances on printed wiring board	- Printed wiring material - Creepage distance - Reinforced insulation	WG 4	2015 CTL Plenary Meeting

Question

1) In case of components mounted on a printed wiring board (PWB) and placed across (bridging) insulating distances on the PWB (e.g. providing basic or even reinforced insulation), how to deal with the different distance requirements for distances on the printed wiring material and for distances on other plastics e.g. the housing material of the component (depending on the applicable material group)?
2) In case of components mounted on a printed wiring board (PWB) and placed across insulating distances on the PWB (e.g. providing basic or even reinforced insulation), how to deal with the different distance requirements, if the material group for PWB and for the components placed on it is not the same (e.g. PWB of material group I and relay base of material group IIIa)?

Decision

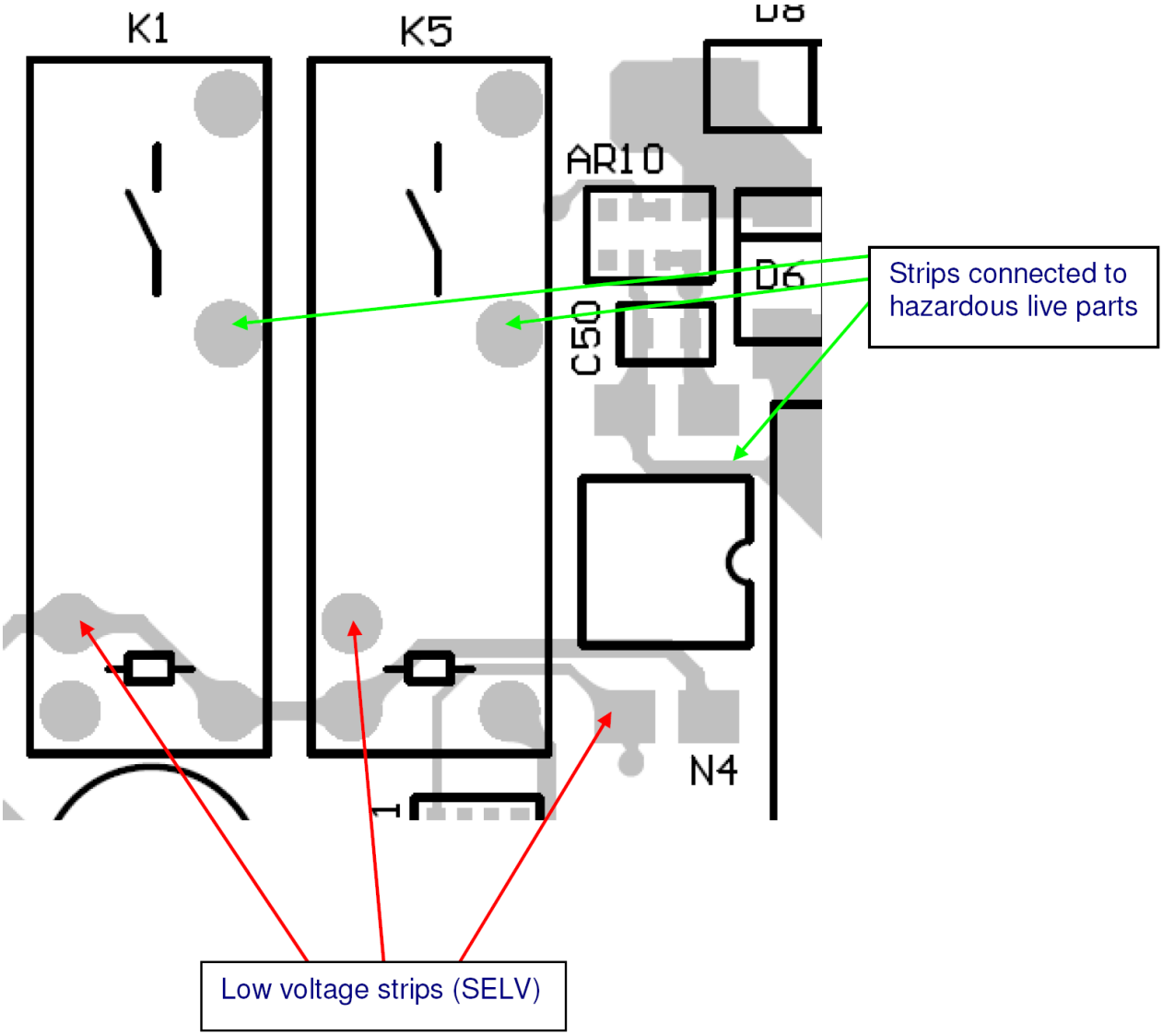
1) The defined reduced values for the required creepage distances relevant to printed wiring material in micro-environment of pollution degrees 1 and 2 may only be applied when there are no components bridging the investigated insulation. If components are bridging insulating distances on the PWB, the creepage distance requirements relating to the (component) material representing the worst case has to be applied for the PWB also.
2) In case of different material groups for the PWB and the components, again the requirement relating to the material representing the worst case has to be used.

Remark: A component is considered to bridge insulation on the PWB, if the surface of the component is located at a distance less than the value “X” (e.g. according to IEC 60664 series) from the isolating PWB surface. See also enclosed drawing.

Explanatory notes

In many standards requirements relevant to creepage distances for basic insulation are provided, which are applicable to printed wiring material in micro-environment of pollution degrees 1 and 2 and which allow an appreciable reduction of distances compared to the requirements for other materials. There is a lack of information on how to apply the requirements in case of components bridging insulation distances on a PWB and in case of plastics of different material class.

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K1 - K5 - N4: components placed on PWB across reinforced insulation

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